BMP Multiple User Handling with Role Types

Overview

This document explains how multiple users from the same BMP client are handled in Activepieces, including:

- **Shared Organizations**: Users with the same `clientId` automatically share organizations
- **Custom Platform Roles**: Users can be assigned different platform roles (ADMIN, MEMBER, OPERATOR, etc.)
- **SDK Authentication**: Using the `/auto-provision` endpoint for seamless user management

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Same ClientId Organization Sharing

How It Works

When multiple users sign up or sign in with the **same `clientId` but different email addresses**, they will **automatically share the same organization**.

Flow Example

First User (User A):

1. Provides `clientId: "client-123"` and `clientName: "MyCompany"`
2. System stores `clientId` in User A's record
3. System creates organization "MYCOMPANY"
4. User A is assigned to this organization

Second User (User B):

1. Provides `clientId: "client-123"` (same as User A)
2. System stores `clientId` in User B's record
3. System checks: "Do any other users with `clientId: "client-123"` already have an organization?"
4. **YES** - User A already has organization "MYCOMPANY"
5. System assigns User B to the **same organization**
6. User B joins "MYCOMPANY" with User A

Implementation Details

```
// After storing clientId
const usersWithSameClientId = await userService.getByClientIdAndPlatform({
    clientId: params.clientId,
    platformId: params.platformId,
})

// Find existing organization from another user with same clientId
const existingOrgUser = usersWithSameClientId.find(
    u => u.id !== user.id && u.organizationId
)

if (existingOrgUser && existingOrgUser.organizationId) {
    // Assign new user to existing organization
    await userService.update({
        id: user.id,
        platformId: params.platformId,
        organizationId: existingOrgUser.organizationId,
    })
}
```

Important Notes

1. **Organization Name:** The organization name is determined by the **first user** who provides `clientName`. Subsequent users join that organization regardless of their `clientName` value.
2. **No `clientName` Provided:** If a user signs in without `clientName` but another user with the same `clientId` already has an organization, the user will be assigned to that existing organization.
3. **Platform Isolation:** Users with the same `clientId` only share organizations within the same `platformId`. Different platforms have separate organizations.

Platform Roles

Available Roles

Role	Description	Permissions
SUPER_ADMIN	System-wide administrator	Full system access
OWNER	Platform/tenant owner	Full platform access
ADMIN	Platform administrator	Can create projects, manage users
MEMBER	Regular platform member	Limited permissions, needs invites
OPERATOR	Platform operator	Automatic project access (except private)

Role Assignment Logic

Standard Signup (`/sign-up`)

- **With invitation**: Uses role from invitation (or defaults to MEMBER)

- **No invitation**: Defaults to **MEMBER**

Standard Login (`/sign-in`)

- **No role change** - User keeps existing platform role

SDK Authentication (`/auto-provision`)

- **Signup**: Assigns **ADMIN** role (or custom role via `roleName`)
- **Login**: User keeps existing role

Custom Role Assignment

The `roleName` Parameter

The `/auto-provision` endpoint supports custom platform roles via the **`roleName`** parameter.

Parameter Details:

- **Source**: `localStorage.getItem('ada.roleName')` from BMP frontend
- **Type**: String (case-insensitive)
- **Valid Values**: `SUPER_ADMIN` , `OWNER` , `ADMIN` , `MEMBER` , `OPERATOR`
- **Default**: `ADMIN` (if not provided or invalid)

Role Mapping Function

```
function mapRoleNameToPlatformRole(roleName?: string): PlatformRole {
    if (!roleName) {
        return PlatformRole.ADMIN // Default role
    }

    const normalizedRoleName = roleName.toUpperCase().trim()

    switch (normalizedRoleName) {
        case 'SUPER_ADMIN': return PlatformRole.SUPER_ADMIN
        case 'OWNER': return PlatformRole.OWNER
        case 'ADMIN': return PlatformRole.ADMIN
        case 'MEMBER': return PlatformRole.MEMBER
        case 'OPERATOR': return PlatformRole.OPERATOR
        default: return PlatformRole.ADMIN // Invalid role, default to ADMIN
    }
}
```

When Role is Applied

Signup Flow:

1. Creating auto-invitation → Uses `platformRole` from `roleName`
2. Creating user directly → Assigns `platformRole` from `roleName`
3. Defaults to ADMIN if `roleName` not provided

Login Flow:

- Role is **NOT changed** during login
- User keeps their existing role from signup

Important Notes

1. **Case Insensitive**: `"admin"`, `"ADMIN"`, `"Admin"` all map to `PlatformRole.ADMIN`
2. **Default Behavior**: Invalid or missing `roleName` defaults to `ADMIN` for backward compatibility
3. **Role Persistence**: Role is set during signup only; login does NOT change the user's role
4. **Security**: Role assignment happens server-side (client cannot bypass validation)

BMP Frontend Integration

JavaScript Example

```
// Get role and client info from localStorage
const roleName = localStorage.getItem('ada.roleName')      // e.g., "ADMIN", "MEMBER"
const clientName = localStorage.getItem('ada.clientName')  // e.g., "MyCompany"

// Prepare auto-provision request
const requestBody = {
  email: userEmail,
  password: userPassword,
  firstName: userFirstName,
  lastName: userLastName,
  platformId: '2Y6xAoWbvjiBgdRsBDcbP',  // Your platform ID
  clientId: 'client-123',                // Unique client identifier
  clientName: clientName,                // Organization name (optional)
  roleName: roleName                     // Custom role (optional)
}

// Call auto-provision endpoint
const response = await fetch('http://localhost:3000/api/v1/authentication/auto-provision', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify(requestBody)
})

const data = await response.json()
// Store token and user info
localStorage.setItem('token', data.token)
```

Setting Role in BMP Frontend

```
// Set role before signup/login
localStorage.setItem('ada.roleName', 'MEMBER')    // For regular users
localStorage.setItem('ada.roleName', 'ADMIN')     // For administrators
localStorage.setItem('ada.roleName', 'OPERATOR')  // For operators
```

Auth Data Flow Diagrams

The React UI SDK does **not** authenticate users itself; the host application supplies the token and config (`apiUrl` , `token` , `projectId`). Two flows are described below: the **original SDK flow** (generic host passing config to the SDK) and the **bmp-fe-web flow** (host uses `/auto-provision` and then passes config to the SDK). Implementation reference: `bmp-fe-web/src/app/pages/activepieces/page/activepieces.component.ts` .

Role of platformId

- **bmp-fe-web:** `platformId` comes from the host **environment** (e.g. `environment.activepieces?.platformId`). It scopes auto-provision to a tenant (together with `clientId`), and when set it triggers `ensureOrganizationSetup` after auth.
- **Activepieces (backend): platformId is required on the Activepieces side for multi-tenant handling.** Each platform is a tenant; user, project, and organization are scoped by `platformId` . For SDK/embedded use (e.g. one host like bmp-fe-web serving multiple tenants), the client must send `platformId` in the request body so Activepieces knows which tenant to use; otherwise the API may return a validation error or resolve to the wrong tenant.

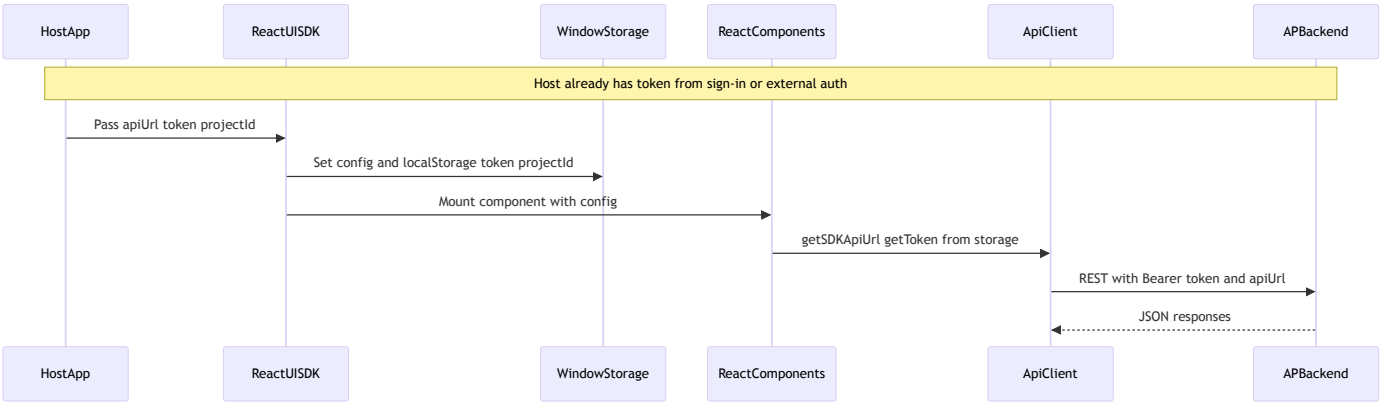
Importance of platformId for the SDK config

The **SDK config** (what the host passes to the SDK) is `apiUrl` , `token` , and `projectId` (optional `flowId`). The SDK does **not** receive `platformId` as a config field; it only uses `apiUrl` , `token` , and `projectId` for API calls.

1. **platformId is required on the Activepieces side for multi-tenant handling** — each platform is a tenant; the backend needs `platformId` to scope the request and to create/return the correct token and projectId.
2. The host must set the correct `platformId` (e.g. in environment) when calling auto-provision so the resulting SDK config is for the intended tenant; wrong or missing `platformId` yields wrong or failed SDK auth.

Diagram 1: Original SDK Authentication Flow

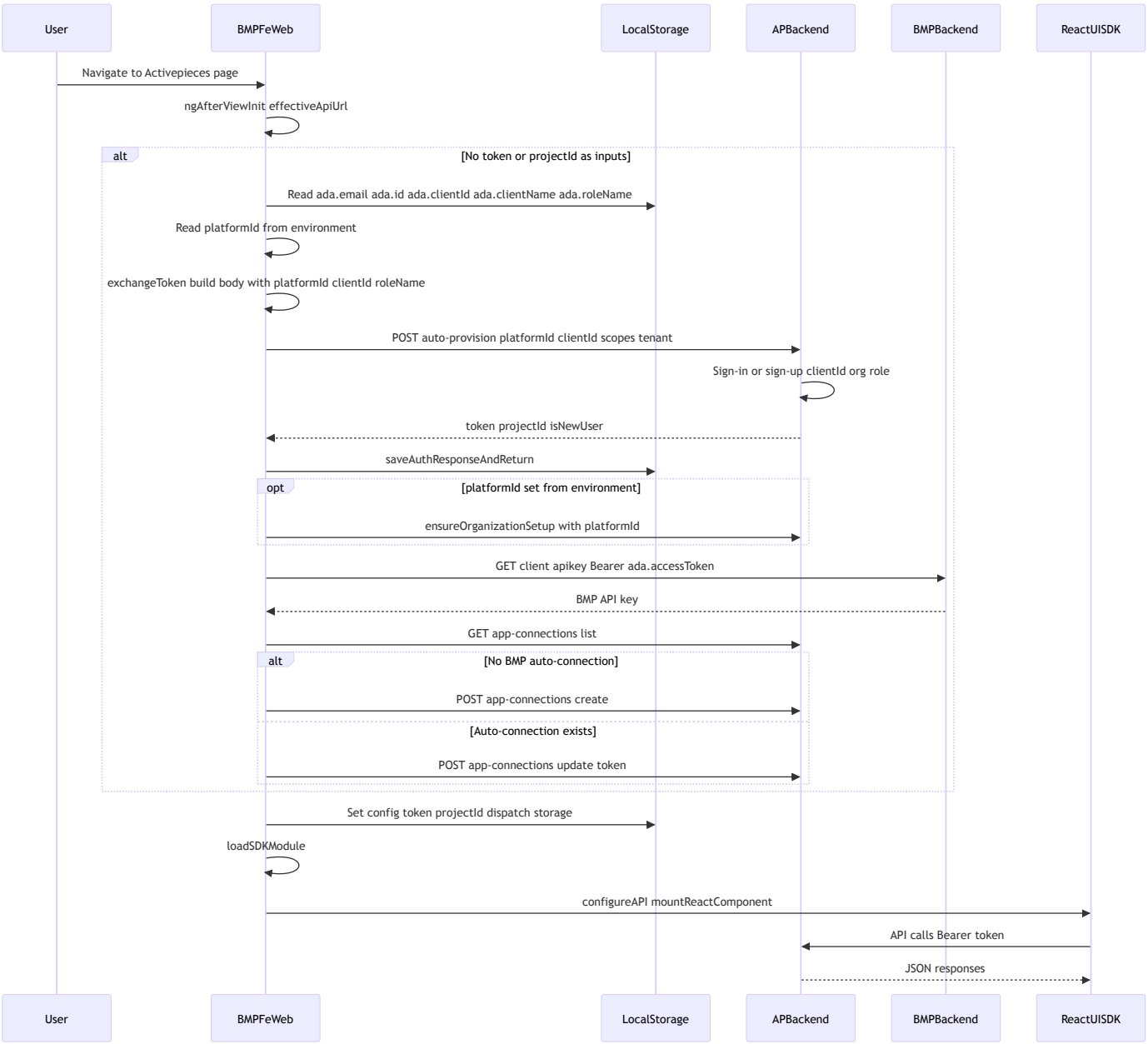
The SDK does not perform login. The host application obtains a token (e.g. its own login or standard Activepieces sign-in) and passes `apiUrl` , `token` , and `projectId` into the SDK; the SDK only stores and uses them for API calls.



Caption: Host obtains token (e.g. standard sign-in/sign-up or its own auth). Host passes `apiUrl` , `token` , `projectId` to the SDK via Angular inputs or `SDKProviders` config. SDK writes these to `window.__AP_SDK_CONFIG__` and to localStorage. React UI (and `api.ts`) read config and token from storage for all API requests; the SDK never calls sign-in or sign-up itself. When the host uses platform-scoped auth such as auto-provision, the correct `platformId` in the host environment is what ensures the token and projectId passed to the SDK are for the right tenant (see "Importance of platformId for the SDK config" above).

Diagram 2: bmp-fe-web Authentication Flow

platformId is required on the Activepieces side for multi-tenant handling: each platform is a tenant; the backend needs `platformId` to determine which tenant to use and will return a validation error if it cannot be determined. The sequence below matches `bmp-fe-web` (ActivepiecesComponent): `ngAfterViewInit` → `exchangeToken` (localStorage ada.* and platformId from environment) → `ensureOrganizationSetup` when platformId is set → `ensureBmpConnection` → set config → load SDK → mount component.



Caption: bmp-fe-web reads `ada.email`, `ada.id` (used as password), `ada.clientId`, `ada.clientName`, and `ada.roleName` from localStorage, and **platformId** from **environment** (`environment.activepieces.platformId`). It POSTs to `/v1/authentication/auto-provision` with platformId and clientId so the backend scopes the user to that tenant (same platformId + same clientId → same organization). Backend signs in or signs up, applies clientId/organization and role, returns token and projectId. Frontend stores them. If platformId is set, it runs `ensureOrganizationSetup` (org + env with ADA_BMP_API_URL) using that platformId; then `ensureBmpConnection` (GET BMP API key from BMP backend, then create/update BMP app-connection via Activepieces API). Finally it sets `__AP_SDK_CONFIG__`, loads the SDK script, calls `configureAPI` and `mountReactComponent`. From then on, SDK behavior matches the original SDK flow (Diagram 1).

API Examples

Example 1: Team Admin with Custom Organization

```
POST /api/v1/authentication/auto-provision
{
  "email": "admin@company.com",
  "password": "secure123",
  "firstName": "Admin",
  "lastName": "User",
  "platformId": "2Y6xAoWbvjiBgdRsBDcbP",
  "clientId": "company-abc-123",
  "clientName": "MyCompany",
  "roleName": "ADMIN"
}

// Result: User created with ADMIN role in "MYCOMPANY" organization
```

Example 2: Team Member Joining Existing Organization

```
POST /api/v1/authentication/auto-provision
{
  "email": "member@company.com",
  "password": "secure456",
  "firstName": "Team",
  "lastName": "Member",
  "platformId": "2Y6xAoWbvjiBgdRsBDcbP",
  "clientId": "company-abc-123",      // Same clientId as admin
  "clientName": "MyCompany",
  "roleName": "MEMBER"
}

// Result: User created with MEMBER role, automatically joins "MYCOMPANY" organization
```

Example 3: Without Custom Role (Default ADMIN)

```
POST /api/v1/authentication/auto-provision
{
  "email": "user@example.com",
  "password": "password123",
  "firstName": "John",
  "lastName": "Doe",
  "platformId": "2Y6xAoWbvjiBgdRsBDcbP",
  "clientId": "client-789",
  "clientName": "JohnOrg"
  // roleName not provided → defaults to ADMIN
}

// Result: User created with ADMIN role in "JOHNORG" organization
```

Example 4: Operator Role

```
POST /api/v1/authentication/auto-provision
{
  "email": "operator@company.com",
  "password": "secure789",
  "firstName": "Operator",
  "lastName": "User",
  "platformId": "2Y6xAoWbvjiBgdRsBDcbP",
  "clientId": "company-abc-123",      // Same clientId as others
  "clientName": "MyCompany",
  "roleName": "OPERATOR"
}

// Result: User created with OPERATOR role, joins "MYCOMPANY" organization
//         Has automatic access to non-private projects
```

Use Cases

Use Case 1: Company with Multiple Team Members

Scenario: A company has an admin, several members, and operators.

Setup:

```
// 1. Admin signs up first
{
  email: "admin@company.com",
  clientId: "company-123",
  clientName: "CompanyName",
  roleName: "ADMIN"
}

// 2. Team members join
{
  email: "member1@company.com",
  clientId: "company-123",  // Same clientId
  roleName: "MEMBER"
}

{
  email: "member2@company.com",
  clientId: "company-123",  // Same clientId
  roleName: "MEMBER"
}

// 3. Operator joins
{
  email: "ops@company.com",
  clientId: "company-123",  // Same clientId
  roleName: "OPERATOR"
}
```

Result:

- ✅ All users share the "COMPANYNAME" organization
- ✅ Admin can create and manage projects
- ✅ Members need project invites
- ✅ Operator has automatic access to projects

Use Case 2: Multi-Client SaaS Platform

Scenario: BMP frontend serves multiple client companies.

Client A:

```
{
  email: "user1@clienta.com",
  clientId: "client-a",
  clientName: "Client A Corp",
  roleName: "ADMIN"
}
```

Client B:

```
{
  email: "user1@clientb.com",
  clientId: "client-b",      // Different clientId
  clientName: "Client B Inc",
  roleName: "ADMIN"
}
```

Result:

- ✅ Client A users share one organization
- ✅ Client B users share a separate organization
- ✅ Complete isolation between clients
- ✅ Each can have their own role structure

Use Case 3: Gradual Team Growth

Scenario: Company starts with one user, adds more over time.

Timeline:

1. **Week 1**: First user signs up
   - Creates organization "MYCOMPANY"
   - Gets ADMIN role
2. **Week 2**: Second user joins
   - Same `clientId`
   - Automatically joins "MYCOMPANY"
   - Gets MEMBER role
3. **Week 3**: Third user joins
   - Same `clientId`
   - Automatically joins "MYCOMPANY"
   - Gets OPERATOR role

Result:

- ✅ Seamless team growth
- ✅ No manual organization assignment needed
- ✅ Each user has appropriate role
- ✅ All share same resources

Summary Table

Feature	Description
Endpoint	`/api/v1/authentication/auto-provision`
Organization Sharing	Same `clientId` → Same organization
Default Role	`ADMIN` (if `roleName` not provided)
Custom Roles	Via `roleName` parameter from localStorage
Role Options	SUPER_ADMIN, OWNER, ADMIN, MEMBER, OPERATOR
Platform Isolation	Different `platformId` → Separate organizations
Backward Compatible	✅ Yes - defaults to ADMIN

Database Schema

```
-- User table
user {
    id: string
    email: string (via identity)
    clientId: string (nullable)
    organizationId: string (nullable)
    platformId: string
    platformRole: enum (SUPER_ADMIN, OWNER, ADMIN, MEMBER, OPERATOR)
}

-- Organization table
organization {
    id: string
    name: string (e.g., "MYCOMPANY")
    platformId: string
}
```

Testing Checklist

- ✅ Same `clientId` with different emails → Users share organization
- ✅ Different `clientId` → Separate organizations

- ✅ `roleName: "ADMIN"` → User gets ADMIN role
- ✅ `roleName: "MEMBER"` → User gets MEMBER role
- ✅ `roleName: "OPERATOR"` → User gets OPERATOR role
- ✅ `roleName` not provided → Defaults to ADMIN
- ✅ Invalid `roleName` → Defaults to ADMIN
- ✅ Case-insensitive role names work correctly
- ✅ Login preserves existing role (no change)
- ✅ Platform isolation works (different platformId)
- ✅ New user's project inherits user's `organizationId` automatically
- ✅ Existing identity with new platform user → Project gets correct `organizationId`

Auto-Create Default BMP Connection (Frontend)

The default BMP connection is created by the **BMP frontend** (`bmp-fe-web`) using the `ensureBmpConnection` function in `activepieces.component.ts` .

How It Works

1. After successful authentication, the frontend calls `ensureBmpConnection`
2. It fetches the real BMP API key from `GET /client/apikey`
3. Checks if a BMP connection already exists for the project
4. If an auto-connection exists, **updates the token** to keep it in sync
5. If no connection exists, **creates a new one** with the real API key

Benefits of Frontend Approach

- Uses the **real BMP API key** (not a placeholder)
- **Updates tokens** on each visit to keep them in sync
- Runs in the **user's session context** where BMP credentials are available

Related Files

- Frontend: `bmp-fe-web/src/app/pages/activepieces/page/activepieces.component.ts`
  - `ensureBmpConnection()` - Main function
  - `getBmpApiKey()` - Fetches API key from BMP backend
  - `createBmpConnection()` - Creates new connection
  - `updateBmpConnectionToken()` - Updates existing connection token

Required request body for POST /v1/app-connections (BMP / Custom Auth)

The API expects a **body** that matches the **Custom Auth** schema. A **400 Bad Request** usually means a required field is missing or the shape is wrong.

- **Required body fields** (all must be present):
  - `projectId` (string) – **Must be in the body**; used for authorization. Use the current project ID (e.g. from `/auto-provision` or project context).
  - `type` : `"CUSTOM_AUTH"`
  - `externalId` (string) – e.g. a stable id for the BMP connection
  - `displayName` (string)

    - `pieceName` : `"@activepieces/piece-ada-bmp"`
    - `value` : `{ type: "CUSTOM_AUTH", props: { apiToken: "<BMP API key>", environment: "Dev" | "Staging" | "Production" } }`
- **Optional**: `metadata` , `pieceVersion`

Example (create/upsert BMP connection):

```
{
  "projectId": "<current-project-id>",
  "type": "CUSTOM_AUTH",
  "externalId": "bmp-auto-connection",
  "displayName": "ADA BMP",
  "pieceName": "@activepieces/piece-ada-bmp",
  "value": {
    "type": "CUSTOM_AUTH",
    "props": {
      "apiToken": "<BMP API key>",
      "environment": "Staging"
    }
  }
}
```

Ensure the request also sends the project scope (e.g. `projectId` query param or header if your proxy/API expects it) so the backend can authorize the request.

Benefits

1. **Automatic Team Management** - No manual organization assignment needed
2. **Flexible Role Structure** - Support for various permission levels
3. **Seamless Onboarding** - New users automatically join correct organization
4. **Client Isolation** - Different clients remain completely separate
5. **Backward Compatible** - Existing integrations continue to work
6. **Secure** - Server-side role validation and assignment
7. **Auto BMP Connection** - Default connection created by frontend with real API key

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Related Files

- Backend Implementation: `/packages/server/api/src/app/authentication/authentication.controller.ts`
- User Service: `/packages/server/api/src/app/user/user-service.ts`
- User Entity: `/packages/server/api/src/app/user/user-entity.ts`
- Frontend BMP Connection: `bmp-fe-web/src/app/pages/activepieces/page/activepieces.component.ts`